
Hamiltonian Neural Networks for Chaotic Dynamics: Stability Analysis and Cross-Domain Generalization

Jingjie Qiu

Department of Electrical and Computer Engineering
University of California, San Diego
jiq013@ucsd.edu

Hehua Ouyang

Department of Electrical and Computer Engineering
University of California, San Diego
heouyang@ucsd.edu

Tien-Ning Lee

Department of Electrical and Computer Engineering
University of California, San Diego
till104@ucsd.edu

Vincent Kao

Department of Electrical and Computer Engineering
University of California, San Diego
v1kao@ucsd.edu

1 Problem Background & Motivation

Predicting the long-term dynamics of chaotic physical systems remains a fundamental challenge in computational physics and machine learning. Standard deep learning models are fundamentally data-driven; while they excel at short-term trajectory fitting, they lack an intrinsic understanding of underlying physical laws. In highly nonlinear, energy-conserved environments, accumulated numerical errors inevitably lead to energy drift, causing long-term predictions to diverge completely from the true physical manifold. To systematically address this limitation, we select the 2D Hénon-Heiles (HH) system as our primary testbed. The HH system is a classic model for Hamiltonian chaos featuring a strict escape energy threshold, serving as the ultimate stress test for deep learning architectures.

The primary motivation of our project is to overcome this trajectory divergence by constraining neural networks with fundamental physical laws, specifically Hamiltonian mechanics and global energy conservation. While we leverage the HH system as a rigorous theoretical baseline, our broader vision extends far beyond a single theoretical construct. We aim to bridge the critical gap between purely data-driven algorithms and reliable, long-term predictions in broader classes of complex dynamical systems. By forcing the model to inherently respect underlying physical invariants, our ultimate goal is to develop a robust and universally applicable predictive framework. We seek to demonstrate that this physics-informed approach can ensure structural stability across diverse scientific and engineering domains, where unphysical predictions would otherwise lead to catastrophic system failures.

2 Related Work

We categorize the relevant literature into three main themes based on their methodological approaches:

- **Traditional Numerical Integrators and Dynamical Analysis:** Historically, the Hénon-Heiles system has been studied extensively utilizing high-precision numerical schemes to compute alignment indices (SALI) and classify regular versus chaotic orbital behaviors (1).

While standard numerical solvers are widely utilized, they are pure numerical simulations and lack the ability to learn dynamics directly from high-dimensional observational data or generalize across different initial conditions.

- **Purely Data-Driven Dynamics Learning:** Recent deep learning approaches attempt to learn continuous dynamical systems directly from time-series data using architectures like Neural ODEs. While highly effective for dissipative systems, they struggle with energy-conserved chaotic systems because they do not explicitly encode physical invariants, often generating unphysical trajectories during extended temporal rollouts.
- **Physics-Informed and Hamiltonian Neural Networks:** This category explicitly incorporates physical laws into the optimization process. Hamiltonian Neural Networks (HNNs) parameterize the system’s Hamiltonian rather than the vector field, forcing the network’s output to obey Hamilton’s equations via automatic differentiation (2). Our project builds upon this foundational framework to enhance predictive stability in strongly chaotic regimes.

3 High-level Methodology

Our proposed methodology builds upon the existing HNN framework and is structured into three focused phases:

Phase 1: Baseline Establishment

We initially reconstruct the standard HNN baseline for the HH system. This phase serves to establish a robust foundation for subsequent physical and structural modifications, ensuring that the model correctly approximates the basic Hamiltonian surfaces under low-energy conditions.

Phase 2: Physics-Informed Architectural and Constraint Optimization

This phase constitutes the core technical exploration of our project. We systematically investigate the intersection of network topology and physical regularization to enhance predictive stability. On the architectural front, we propose exploring various structural adaptations to the baseline HNN. Our objective is to identify neural configurations that are fundamentally better equipped to represent complex Hamiltonian vector fields. Concurrently, on the optimization front, we will investigate the integration of broader physics-informed regularization mechanisms. By incorporating generalized physical constraints into the learning process, we aim to penalize non-physical behaviors and force the network to respect conserved quantities. This flexible, dual-track approach allows us to broadly explore the optimal synergy between network design and physical constraints without being restricted to a single structural or mathematical formulation.

Phase 3: Cross-Domain Validation and Application Extension

The final phase aims to validate the generalization capability and universality of our optimized HNN framework beyond the idealized Hénon-Heiles system. Rather than restricting our scope to a single theoretical model, we propose to conduct cross-domain validations on broader classes of complex dynamical systems. By evaluating the model’s performance on other representative Hamiltonian analogs—ranging from classical mechanical systems to advanced engineering control environments—we seek to demonstrate that encoding physical priors (i.e., energy conservation) provides a universal advantage. This phase effectively bridges the gap between theoretical physical models and practical applications, proving the framework’s robustness in preventing unphysical divergence across diverse engineering domains.

References

- [1] E. E. Zotos, “Classifying orbits in the classical Hénon-Heiles Hamiltonian system,” *Nonlinear Dynamics*, vol. 79, pp. 1665–1677, 2015.
- [2] M. Mattheakis, D. Sondak, A. S. Dogra, and P. Protopapas, “Hamiltonian neural networks for solving equations of motion,” *Physical Review E*, vol. 105, no. 6, p. 065305, 2022.